

Chapter 5: Asymmetric Ciphers

I. Prime Numbers

An integer p > 1 is a prime number if and only if its only divisors are 1 and itself
Ex. 2,3,5,7 are prime, 4,6,8,9,10 are not

List of prime numbers from 1 to 200 are:

2 3 5 7 11 13 17 19 23 29 31 37 41 43 47 53 59 61 67 71 73 79 83 89 97 101 103 107 109 113 127 131 137 139 149 151 157 163 167 173 179 181 191 193 197 199

Prime Factorisation:

The prime factorisation of a number n is written as a product of its prime factors
eg. 91=7x13 ; 3600=24x32x52

Relative Prime Number

Two numbers a, b are relatively prime if have no common divisors apart from 1

Ex. 8 & 15 are relatively prime

since factors of 8 are 1,2,4,8 and of 15 are 1,3,5,15 and 1 is the only common factor

II. Euler's Theorem

The Euler's totient function ø(n), defined as the number of positive integers less than n & relatively prime to n.

Ex. for n=10 ø(n) = {1,3,7,9}

Term "residue" refers to numbers less than some modulus, and the "reduced set of residues" to those numbers (residues) which are relatively prime to the modulus (n).

Ex for n=10,



complete set of residues is {0,1,2,3,4,5,6,7,8,9}

Reduced set of residues is {1,3,7,9}

Euler's Totient Function:

To compute ø(n) need to count number of residues to be excluded

In general need prime factorization, but

for p (p prime) ø(p) = p-1

for p.q (p,q prime) ø(pq) =(p-1)x(q-1)

eg.

ø(37) = 36

ø(21) = (3–1)x(7–1) = 2x6 = 12

Euler's Theorems

For every a and n that are relatively prime

aø(n) ≡ 1 (mod n) for any a,n where gcd(a,n)=1

Ex.

a=3; n=10; ø(10)=4;

hence 34 = 81 = 1 mod 10

a=2; n=11; ø(11)=10;



hence 210 = 1024 = 1 mod 11

III. Fermat's Theorem

$a^{p-1}=1(\text{mod } p)$

P is prime and a is a positive integer not divisible by p

$3^{201}\text{mod } 11$

3 is not divisible by 11

Hence $a^{p-1}\text{mod } p$

$3^{(11-1)} = 3^{10}=1\text{mod}11$

$(3^{10})^{20}*(3) \text{ mod } 11$

$=1*3 \text{ mod } 11$

$=3$

IV. Testing for Primality

In cryptography prime numbers have its own importance, but to select any large prime number at random is difficult
Thus, we are faced with the task of determining whether a given large number is prime or not .
There is no simple yet efficient means of accomplishing this task.

Miller-Rabin Algorithm

The algorithm due to Miller and Rabin [2] is typically used to test a large number for primality



TEST (n)
1. Find integers k, q, with k > 0, q odd, so that (n-1 = 2kq);
2. Select a random integer a, 1 < q < n-1;
3. if aq mod n = 1 then return (“inconclusive”);
4. for j = 0 to k – 1 do
5. if a2jq mod n = n – 1 then return (“inconclusive”);
6. return (“composite”);.

Example:

Suppose n =29
Find (n- 1) = 28 = 22(7)= 2kq
Let us try a = 10
107 mod 29 = 17 (its neither 1 nor 28, so continue test)
(107)2 mod 29 = 28 hence test return “in clonclusive”

Now let us apply the test to the composite number
N = 13*17 = 221
Then
(n-1) = 220 = 22(55)= 2kq
Let us try a=5
555mod 221 = 112 (neither 1 nor 220)
(555)2 mod 221 = 168 test return “Composite”

V. Principles Of Public-Key Cryptosystems

- Introduction:
  Asymmetric encryption ia a type of cryptosystem in which we use two different keys for encryption and decryption, which are also known as public and private key. It is also known as public-key encryption.

  Asymmetric encryption converts plain text to ciphertext using one of two keys and an encryption algorithm. Using the paired key and decryption algorithm, plaintext is recovered from the ciphertext.

  Asymmetric encryption can be used for confidentiality, authentication, or both.

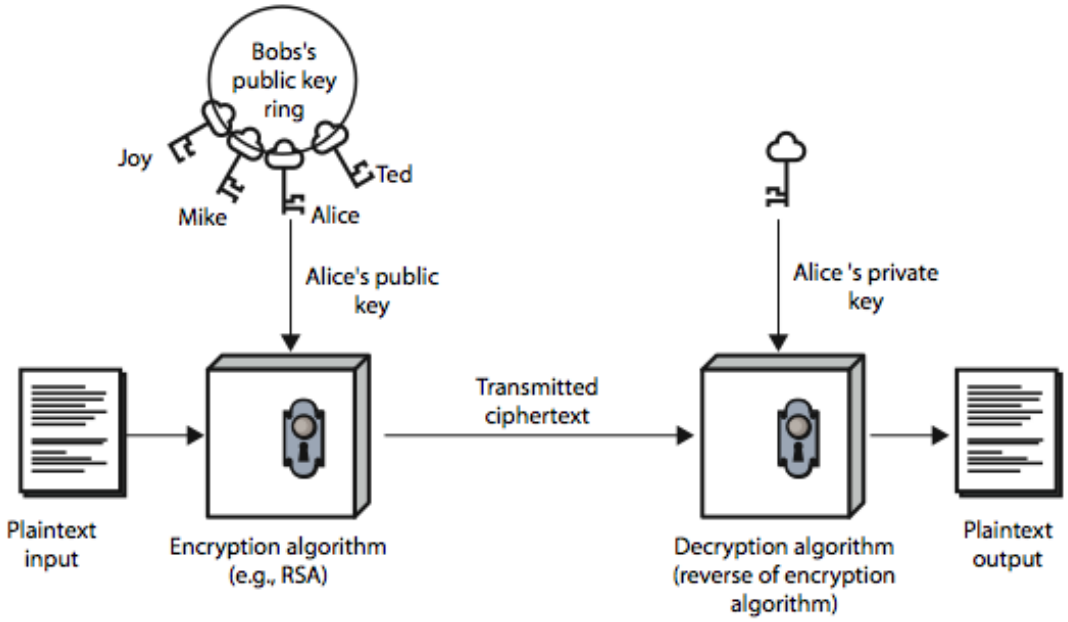


The most widely used public-key cryptosystem is RSA. The difficulty of attacking RSA is based on the difficulty of finding the prime factors of a composite number

- **Principles of Public-key Cryptosystems**.
  - Public-key algorithms are not a substitution or permutation cipher as we learnt in symmetric ciphers but it is based on mathematical functions.
  - Public-key encryption is more secure from cryptanalysis than is symmetric encryption.
  - The concept of public key cryptography was developed while it was tried to attack two of the most difficult problems associated with symmetric encryption.

    (1) Two communicating parties already shared a key, which somehow has been distributed to them.

    (2) The use of key distribution center.
  - The idea of public-key cryptography was invented by Diffie and Hellman.
- **Public-key Cryptosystems**
  - Asymmetric algorithm relies on two different keys, one for encryption and another related key for decryption.
  - Characteristic of PKC: it is computationally infeasible to determine the decryption key given only knowledge of the cryptographic algorithm and encryption key.
  - Public key is known publically, it is infeasible to derived a private key related to that public key.



(a) Encryption

- The essential steps for the communication in PKC are as follows:
- Each user generate a pair of keys to be used for the encryption and decryption of messages.
- Each user places one of the two keys in a public resister or other accessible file. This is public key. The companion key is kept private.
- If bob wishes to send a confidential message to alias, bob encrypts the message using alias's public key.
- When alias receives the message, she decrypt it using her private key. No other recipient can decrypt the message because only alias knows her private key.
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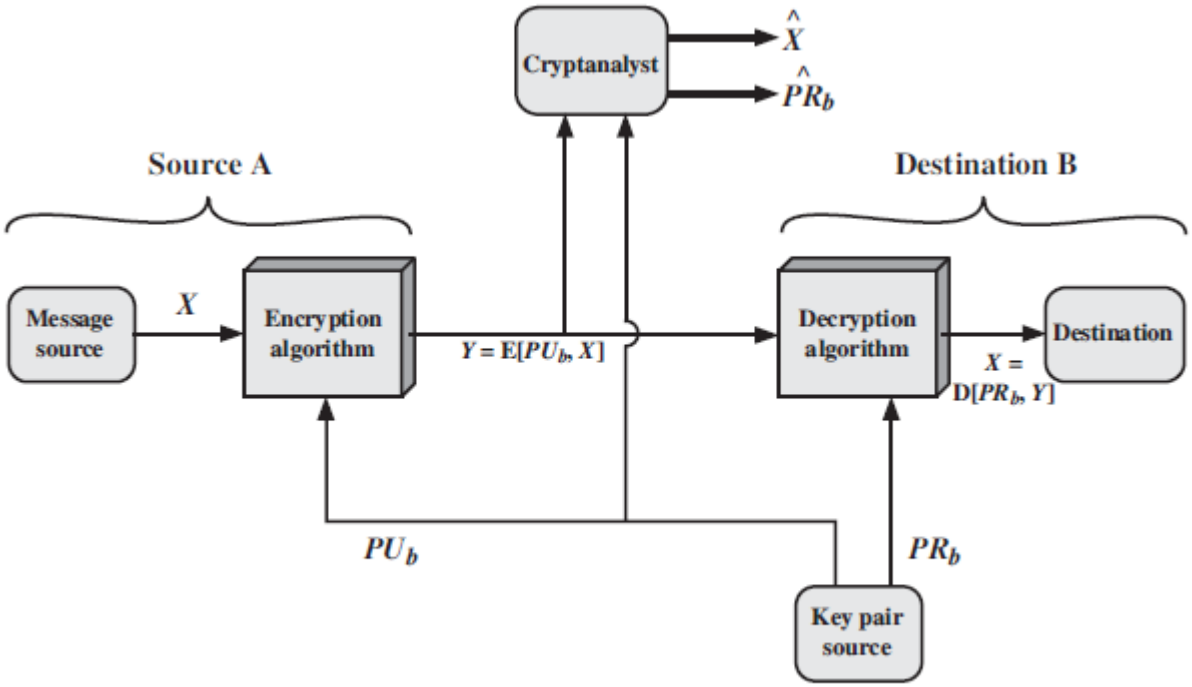


Figure 9.2 Public-Key Cryptosystem: Secrecy

- User A want to send some message to B.
- B generates a related pair of keys: a public key PUb, and private key PRb. PRb is known to only B, whereas PUb is publicly available and so accessible by A.
- With the message X and the encryption key PUb as input, A forms ciphertext y = [y1, y2, y3,….yn]
- Y = E(PUb, X)
- The intended receiver, in possession of the matching private key, is able to invert the transformation:
- X = D(PRb, Y)

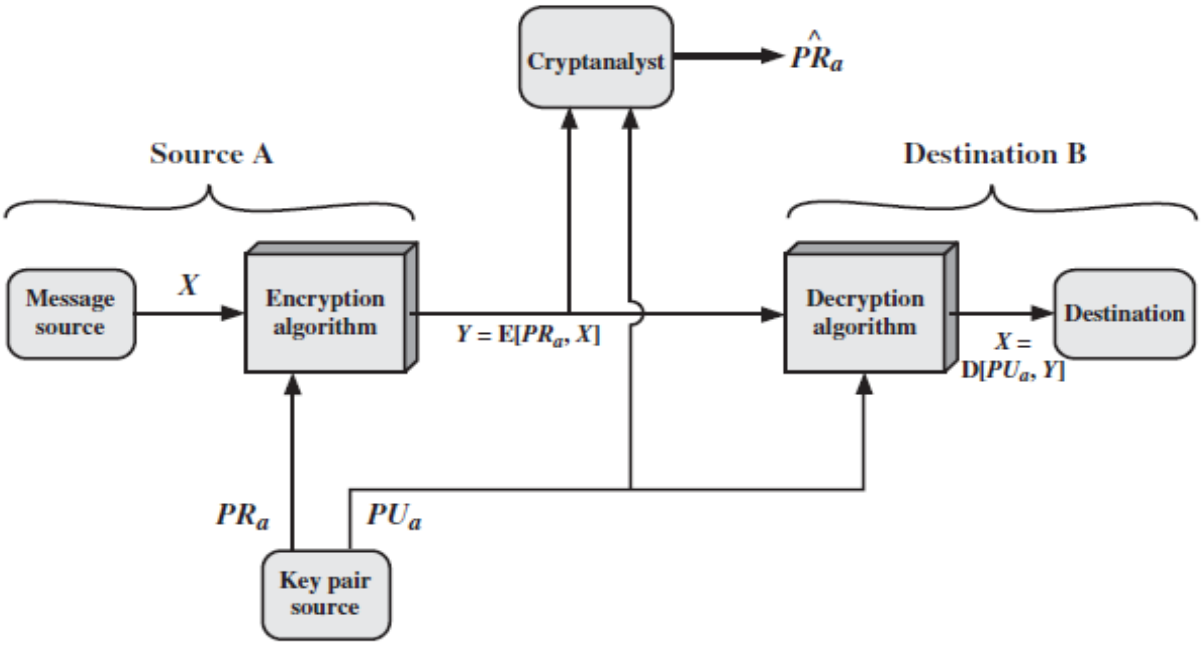


Figure 9.3 Public-Key Cryptosystem: Authentication

- Figurer 9.3 show the use of public-key encryption to provide authentication.
- Y = E(PRa, X)
- X = D(PUa, Y)
- In this case, A prepare a message to B and encrypts it using A's private key before transmitting it.
- B can decrypt it using A's public key .
- Because the message was encrypted using A's private key, only A could have prepare the message.
- So, the entire encrypted message serves as a digital signature.
- In addition, it is impossible to alter the message without access to A's private key, so the message is authenticated both the terms of source and in terms of data integrity.
- In the preceding scheme, the entire message is encrypted, which, although validating both author and contents, requires a great deal of storage.
- Each document must be kept in plaintext to be used for pratical purpose.

- A copy also stored in ciphertext so that the original and contents can be verified in case of dispute.
- A more efficient way of achieving the same results is to encrypt a small block of bits that is a function of the document.
- Such a block, called an authenticator, must have the property that it is infeasible to change the document without without changing the authenticator.
- If the authenticator is encrypted with the sender's private key, it serve as signature that verifies origin, content, and sequencing

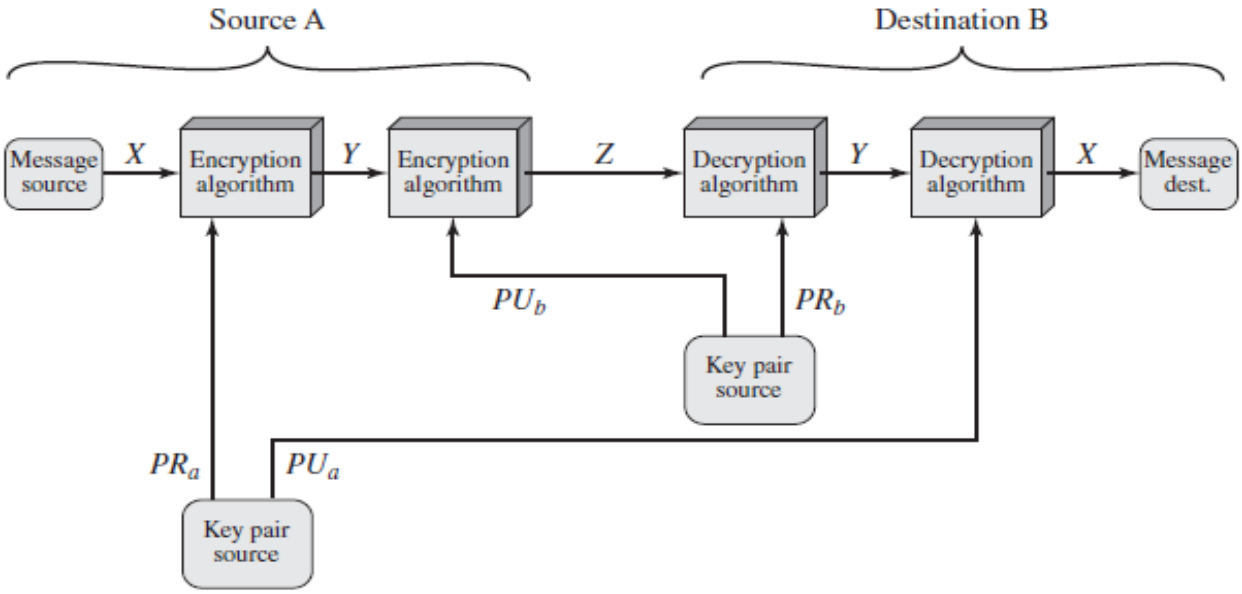


Figure 9.4 Public-Key Cryptosystem: Authentication and Secrecy

- **Applications for Public-key Cryptosystem**
  - In broad terms, we can classify the use of public-key cryptosystem into three categories:
  - Encryption/decryption
  - Digital signature
  - Key exchange
- **Requirements for Public-key Cryptography**
  - The PKC algorithm must fulfill the following conditions



- It is computationally easy for party B to generate the key pair.
- It is computationally easy for a sender A, knowing the public key and the message to be encrypted, M, to generate the corresponding cipher text.
- It is computationally easy for the receiver B to decrypt the resulting cipher text using private key to recover the original message.
- It is computationally infeasible for an adversary, knowing the public key to determine the private key.
- It is computationally infeasible for an adversary, knowing the public key and cipher text to recover the original message

VI. The RSA Algorithm

- RSA - Ron Rivest, Adi Shamir and Len Adleman at MIT, in 1977.
  - RSA is a block cipher
  - The most widely implemented
  - Diffie-Hellman
  - Exchange a secret key securely
  - Compute discrete logarithms
  - Best known & widely used public-key scheme
- Can be used to provide both secrecy & digital signatures
- Based on exponentiation in a finite (Galois) field over integers modulo a prime, using large integers (eg. 1024 bits)
- Uses large integers for key sizes typically 1024 bits

Key Generation Alice

- Select p,q p and q both are prime, p≠q
- Calculate n= p*q
- Calculate ø (n)= (p-1)*(q-1)
- Select integer e gcd(ø(n),e)=1; 1<e< ø(n)
- Calculate d $d=e^{-1}$ (mod(ø(n)))
- Public Key PU= {e,n}
  Private Key PR={d,n}

Encryption by Bob with Alice's Public Key



- Plaintext: M<n
- Cipher Text C= M^e mod n

Decryption by Alice with Alice's Private Key

- Cipher text: C
- Plain Text: M= C^d mod n

- To encrypt a message

  c = m^e mod n

- To decrypt a message

  m = c^d mod n

  = $(m^e \text{ mod } n)^d$ mod n

  = m^{ed} mod n

  = $m^{1+k\text{ø}(n)}$mod n

  = m . $m^{k\text{ø}(n)}$mod n

  = m

RSA Example- Key setup

- Select two prime numbers
  - P=17 and q=11
- Compute n

  n= p*q = 17*11= 187
- Compute ø Value

  Ø (n) = (p-1)*(q-1)= (17-1)*(11-1)= 16*10 = 160
- Select Encryption Parameter

  e : gcd(e,160)=1; choose e =7
- Determine Decryption Parameter

  d: de=1 mod 160 and d < 160

  Value of d = 23 since 23*7 = 161
  - Public Key PU= {7, 187}



- Private Key PR= {23, 187}

RSA Example- En/Decryption

- Encrypt message M =88

  C = 88^7 mod 187

  = 88 $^{(3+3+1)}$ mod 187

  = ((88^3 mod 187) (88^3mod 187)(88 mod 187)) mod 187

  = (44 * 44 * 88) mod 187

  = 11

- Decryption is:

  M = 11^{23} mod 187 = 88

RSA Example-2- Key setup

- Select two prime numbers

  P=11 and q=3

- Compute n

  n= p*q = 11*3= 33

- Compute ø Value

  Ø (n) = (p-1)*(q-1)= (11-1)*(3-1)= 10*2 = 20

- Select Encryption Parameter

  e : gcd(e,20)=1; choose e =3

- Determine decryption parameter

  d: de=1 mod 20 and d<20 ; d=7 since

- The sample RSA private/public operations are:
  - Given message M = 7
- Encryption is:
  - C= 7^3 mod 33



- = 343 mod 33
- = 13

- Decryption is:

  $M = 13^7 \bmod 33$

  $= 13^{(3+3+1)} \bmod 33$

  $=((13^3 \bmod 3)*(13^3 \bmod 33)*(13 \bmod 33)) \bmod 33$

  $=19*19*13 \bmod 33 = 4693 \bmod 33 = 7$

- **Consider the text grouping in the groups of three i.e**
  - ATTACKXATXSEVEN = ATT ACK XAT XSE VEN
  - Represent the blocks in base 26 using A=0, B=1, C=2 …
    - $ATT = 0*26^2 + 19*26^1 + 19 = 513$
    - $ACK = 0*26^2 + 2*26^1 + 10 = 62$
    - $XAT = 23*26^2 + 0*26^1 + 19 = 15567$
    - $XSE = 23*26^2 + 18*26^1 + 4 = 16020$
    - $VEN = 21*26^2 + 4*26^1 + 13 = 14313$
  - What should be the value of n?
  - The value of n should be greater than 17575. How & Why?
  - Let p = 137 and q = 131; so that n= pq = 17947
  - Compute phi value:
    - Ø(n) = (p-1)(q-1) = 136*130 = 17680
  - Select encryption parameter
    - e: gcd(e,17680)=1; choose e=3
  - Determine decryption parameter
    - d: de=1 mod 17680 and d< 17680
  - Value is d=11787 since 11787*3 = 35361= 11787*2+1



- Publish public key PU= {3,17680}
- Keep secret private key PR={7,17680}

- The sample RSA private/public operations are:
  - Given message M = ATT = 513
- Encryption is
  - $C = 513^3 \bmod 17947 = 8363$
- Decryption is
  - $M = 8363^{11787} \bmod 17947 = 513$
- Plaintext is represented as the set of integer's m

  {513, 62, 15567, 16020, 14313}
- Cipher text is represented as the set of integers m

  { 8363, 5017, 11884, 9546, 13366}

VII. The Security of RSA

- Four possible approaches to attacking the RSA algorithm are as follows:
- Brute force: this involves trying all possible private keys.
- The defense against the brute-force approach is same for RSA as for other cryptosystem, namely, use a large key space.
- So the large number of bit in d is better.
- However, because the calculations involve, both in key generation and in encryption/decryption, are complex, the large the size of key, the slower the system will run.
- Timing attacks: these depend on the running time of the decryption algorithm.
- Chosen ciphertext attacks: this type of attack exploits properties of the RSA algorithm



- The Factoring Problem: we can identify three approaches to attacking RSA mathematically.
- Factor n into its two prime factors. This enables calculation of ø(n)=(p-1)(q-1), which, in turn, enables determination of $d = e^{-1} \bmod ø(n)$
- Determine ø(n) directly, without first determining p and q. Again, this enables determination of $d = e^{-1} \bmod ø(n)$.
- Determine d directly, without determining ø(n)
- **Timing Attacks:**
    - Paul kocher, a cryptographic consultant, demonstrated that a snooper can determine a private key by keeping track of how long a computer takes to decipher message.
    - Timing attacks are applicable not just for RSA, but to other public-key cryptography systems.
    - This attack is alarming for two reasons: it comes from a completely unexpected direction and it is a ciphertext-only attack.
    - There are simple countermeasures that can be used for timing attack:

      Constant exponentiation time:
    - Ensure that all exponentiations take the same amount of time before returning a result. This is a simple but does degrade performance.

      Random delay:
    - Better performance could be achieved by adding a random delay to the exponentiation algorithm to confuse the timing attack.
    - Blinding: multiply the ciphertext by a random number before performing exponentiation .
    - This process prevents the attacker from knowing what ciphertext bits are being processed inside the computer and so prevents the bit-by-bit analysis essential to the timing attack



VIII. Diffie Hellman algorithm

- Discovered by Whitfield Diffie and Martin Hellman
- “New Directions in Cryptography”
- Diffie-Hellman key agreement protocol
- Exponential key agreement
- Allows two users to exchange a secret key
- Requires no prior secrets
- Real-time over an untrusted network

Algorithm:

- Global Public Elements
  - q prime number
  - α $\alpha < q$ and α a primitive root of q
- User A Key Generation
  - Select private X_A $X_A < q$
  - Calculate public Y_A $Y_A = \alpha^{XA} \bmod q$
- User B Key Generation
  - Select private X_B $X_B < q$
  - Calculate public Y_B $Y_B = \alpha^{XB} \bmod q$
- Calculation of Secret Key by User A

  $K = (Y_B)^{XA} \bmod q$
- Calculation of Secret Key by User B

  $K = (Y_A)^{XB} \bmod q$



Example:

- Alice and Bob get public numbers
    - P = 23, G = 9
    - Alice and Bob choose Private Number
    - XA = 4
    - XB = 3
- Alice and Bob compute public values
    - YA = 94 mod 23 = 6561 mod 23 = 6
    - YB = 93 mod 23 = 729 mod 23 = 16

Alice and Bob exchange public numbers

- Alice and Bob compute symmetric keys
    - K = (YA)^(XB) mod p = 164 mod 23 = 9
    - K = (YB)^(XA) mod p = 63 mod 23 = 9
    - Alice and Bob now can talk securely!

IX. Applications of Diffie Hellman Key Exchange Algorithm :

- Diffie-Hellman is currently used in many protocols, namely:
- Secure Sockets Layer (SSL)/Transport Layer Security (TLS)
- Secure Shell (SSH)
- Internet Protocol Security (IPSec)
- Public Key Infrastructure (PKI)